

Modern tools for hopper design

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Although there are well-established methodologies for hopper design, many process engineers are uncertain as to how to measure powders in the prescribed way, extract the necessary parameters from recorded data, and successfully apply them. As a result, hopper design and the powder testing associated with it is often outsourced to specialists. This incurs significant cost and undermines the operating company's ability to troubleshoot and re-use or retrofit equipment for alternative materials or applications.

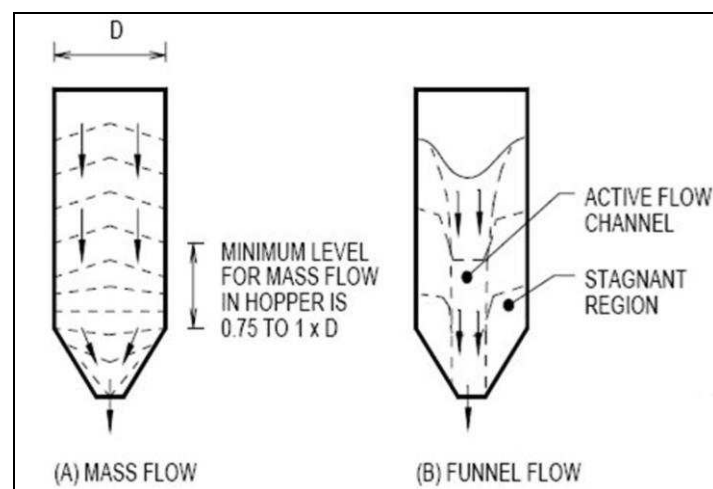
The advent of modern, automated powder testing systems has simplified measurement, making it easier for users who are not instrument specialists to precisely determine the parameters required for hopper design. Furthermore, software developed for the FT4 Powder Rheometer (Freeman Technology) guides engineers through the established methodologies, showing how to use measurements to generate a specification that will operate in an acceptable way. Together these developments bring hopper design easily within the remit of the majority of process engineers, offering an opportunity to reduce costs and gain a better understanding of the factors influencing performance.

Here we examine the issues surrounding hopper design and show, using a worked example, how modern software simplifies the whole process – from powder measurement through to specification.

The basics of hopper design

In this paper the term 'bin' refers to the section of a storage vessel with parallel sided walls, 'hopper' is the angled portion below. A storage vessel or silo therefore consists of both bin and hopper. Many different shapes of hopper and bin are routinely used but in each case the design intent is the same: reliable, steady powder discharge, at the required rate. Selecting an appropriate outlet size and hopper half angle, the degree of incline from the vertical, of the hopper walls, achieves this aim. The resultant flow regimes can be broadly divided into two – mass flow and core or funnel flow.

Figure 1: Flow regimes for powders discharging from a storage silo



Half angle strongly influences the flow mode or regime that develops within the silo (see figure 1). With mass flow (the preferred option for the majority of applications) all of the powder is in motion as material is withdrawn at the exit, producing a 'first in, first out' regime. Flow tends to be relatively consistent and the full capacity of the bin is used. With funnel flow, on the other hand, there is an active channel down the centre of the vessel but powder stagnates along the hopper and bin walls. Steeper hopper walls – smaller hopper half angles – encourage mass as oppose to funnel flow.

Funnel flow produces 'last in, first out' powder delivery and a greater likelihood of operational problems such as rat holing, segregation and flooding. Rat holing is where a central void develops above the discharge outlet in place of the active flow channel. The collapse of rat holes can cause significant mechanical damage and/or excessive aeration of the powder. More generally, aeration in the active flow channel encourages flooding (where the powder becomes fluid-like and flows uncontrollably) and segregation (the separation of particles on the basis of size), both of which are undesirable. While these operational disadvantages discourage the use of funnel flow it can be the preferred choice when building height is limited for example. Funnel flow designs can be short and wide, because the hopper sides are shallowly angled while mass flow units accommodating an equivalent volume tend to be taller with a smaller cross-sectional area.

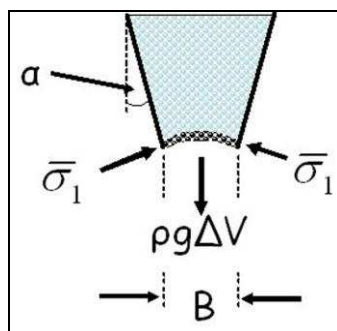
Design theory

Powder flow behaviour in a bin and hopper is governed by:

- The shear properties of the powder – how easily the particles move relative to each other
- Wall friction – how easily the powder flows over the inner surface of the container
- Compressibility – how the application of a consolidating stress changes bulk density

These variables define how the powder will behave in the hopper when consolidated by the weight of material in the bin. Potentially a stable arch can form across the hopper outlet (figure 2), and if this is strong enough to support the rest of the powder in the vessel then discharge ceases. For any given combination of powder and material of construction, hopper half angle and outlet size determine whether a stable arch can form. In the 1960's Jenike carried out a detailed flow analysis based on this flow/no flow condition, to develop a design methodology [1] which remains the standard.

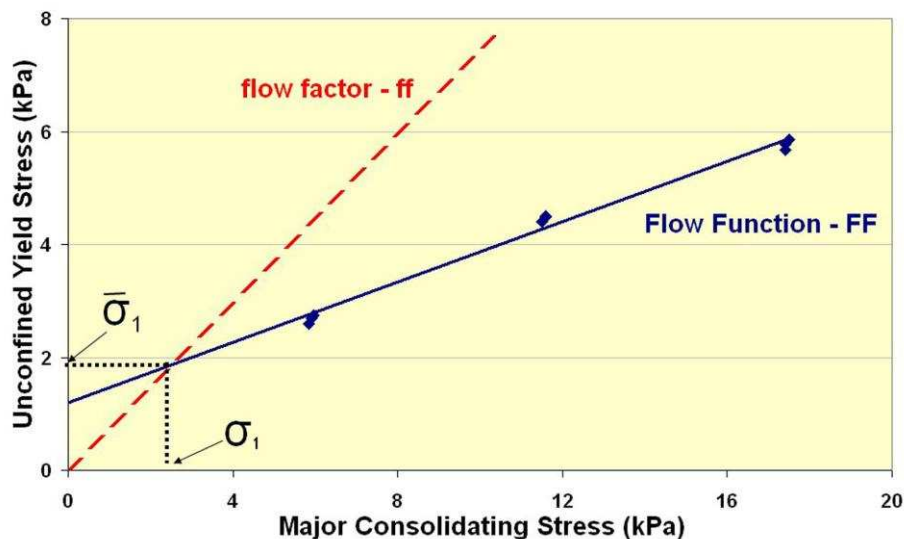
Figure 2: The formation of a stable arch that prevents powder flow depends on the relative size of forces acting within the hopper



A full description of the associated mathematical analysis [2] lies beyond the scope of this paper but in summary the technique involves determining two parameters: flow function (FF) and flow factor (ff). FF depends purely on the shear strength of the powder, which is measured as a function of applied normal stress using shear cell apparatus. The torque or force required to shear a consolidated powder bed across a plane is accurately determined to generate yield loci for the material from which FF is derived. Reference 3 describes shear cell testing methodology and the associated Mohr's circle analysis in some detail.

Flow factor, ff, in contrast, depends on the characteristics of the hopper - material of construction, shape - as well as those of the powder, and is, for any specific hopper configuration, a function of hopper half angle, wall friction and material bulk strength. A plot of FF and ff is shown in figure 3. It is clear that both parameters describe relationships between shear strength and consolidating stress, one for the material itself (FF), the other for the material within the specific hopper environment (ff). The point at which these two curves intersect gives the value of stress in a hypothetical arch at the transition point from flow to no flow. Outlet size is calculated from this value through a simple force balance on the arch.

Figure 3: A plot of FF and ff showing the intersect point defining the flow/no flow transition



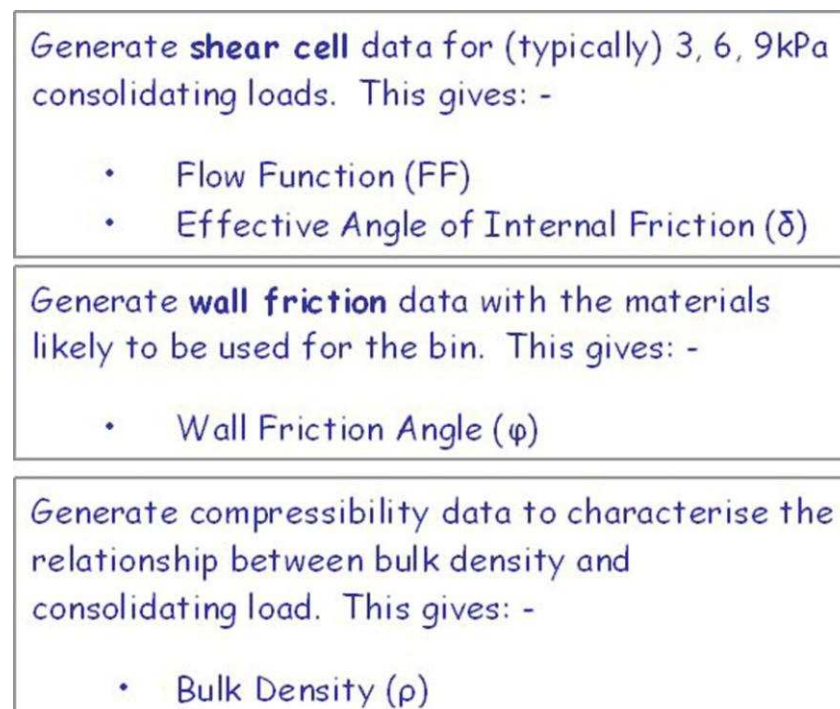
It is important to recognise from this analysis that any change in the FF or ff will alter the critical dimensions of the hopper. If the material of construction, shape or half angle of one hopper is different from that of another, then a different outlet size will be needed to achieve flow, *for the same powder*. If the intent is to use a storage silo for a powder different from the one for which it was designed, then this will alter FF (and ff) and so half angle and outlet size may or may not be adequate. Both these conclusions are fairly obvious. However, what is perhaps less well-understood is that FF and ff may change, for a given material, depending on in-process conditions and the powder properties.

If the material segregates, for example, the hopper may have to cope with slugs of finer and coarser material, which may be more or less cohesive respectively. Moisture level too can cause a significant change in shear strength as can storage time. If the material is allowed to consolidate under its own weight for a significant period then shear strength can rise significantly (time consolidation). Repeated testing under different conditions allows the designer to assess sensitivity to such changes. The choice is then either to specify on the basis of the worst expected case, or install upstream measures to avoid variability that will compromise hopper operation.

Worked example: Determining half angle and outlet diameter for a conical hopper for potato flour

Software developed for the FT4 guides the user through measurement, data work-up and the design methodologies developed by Jenike (figure 4); in this example the design is for potato flour. Automation of every step facilitates precise powder testing and generation of an acceptable design.

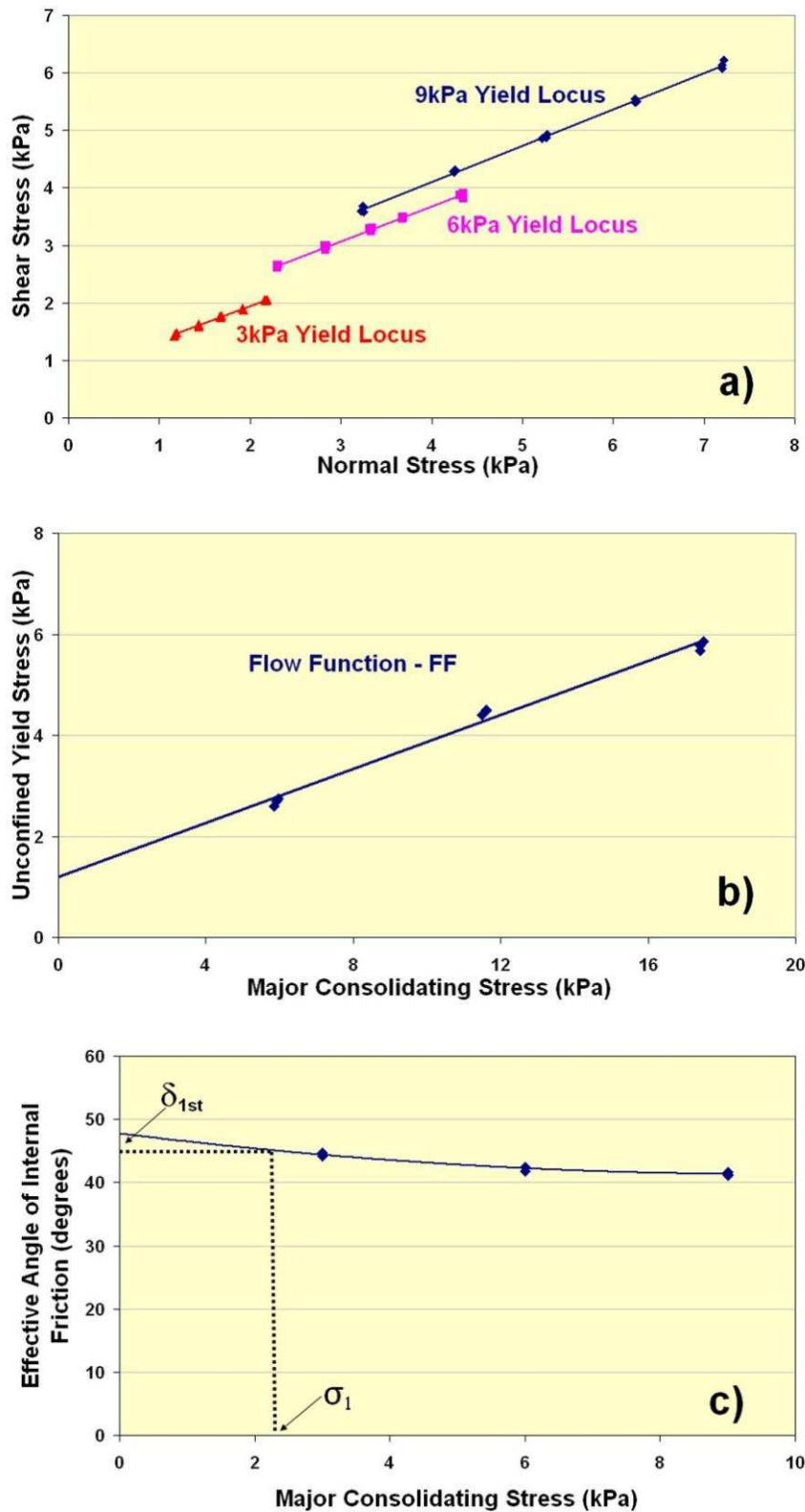
Figure 4: The methodology used to generate critical dimensions for a hopper



Measuring shear data to generate FF and internal angle of friction

During shear cell testing the sample is consolidated at a specified pressure before measuring shear strength as a function of (lower) applied normal stresses to generate a yield locus. Different consolidation pressures are used to generate a series of yield loci from which the FF plot is derived. Figure 5 shows the plots produced by the software for the potato flour, from the shear cell data. The angle of internal friction (AIF), which is used in the calculation of ff, is derived from this test. The AIF is related to consolidating load but in the first iteration of the calculation the stress in the arch is completely unknown so it is common practice to take an average of three measurements for use in subsequent calculations, as an initial approximation. Here this generates a value of 42.7°.

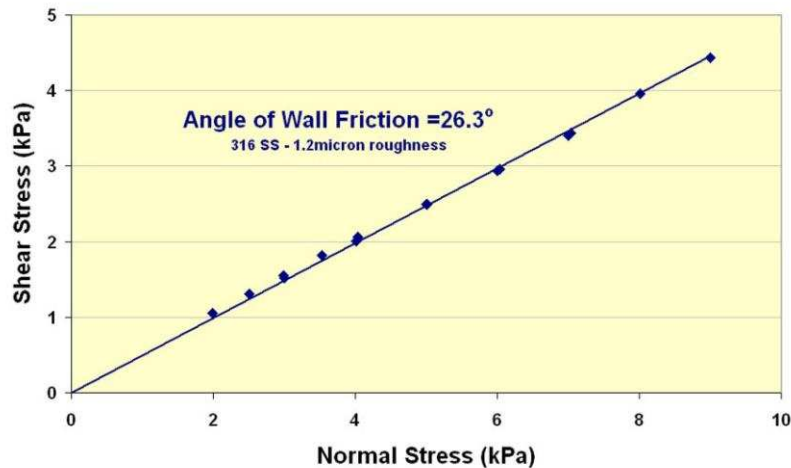
Figure 5: Work up of shear test data a) shear stress as a function of applied normal stress measured at different consolidation loads b) the FF plot c) internal angle of friction as a function of major consolidating stress



Measuring wall friction data to generate wall friction angle

Wall friction is characterised in an analogous way to shear strength by shearing the powder against a surface of the proposed material of construction rather than against itself. For this hopper the intent is to use 316SS with a 1.2 micron surface roughness. Test data generated using the methodologies defined within the instrument software are shown in figure 6 along with the determined value of wall friction angle: 26.3° .

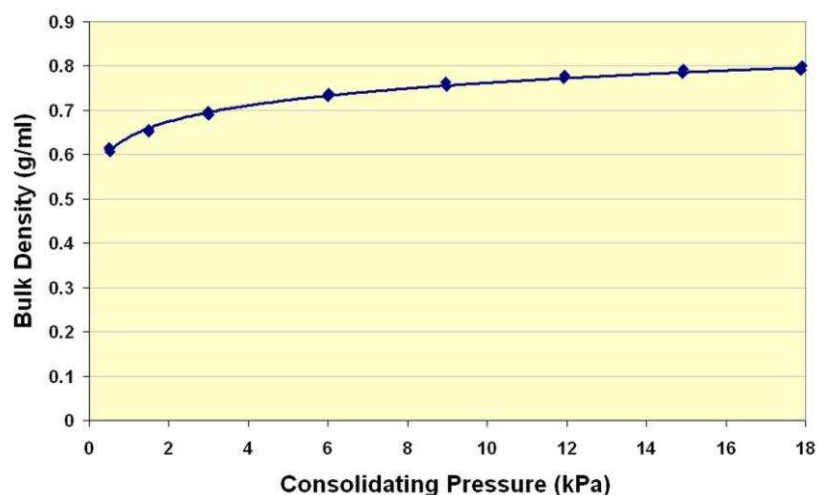
Figure 6: Wall friction test data



Measuring compressibility data to define bulk density as a function of consolidating pressure

The bulk density of a powder can change markedly depending on the degree to which it is compressed. In general more cohesive materials exhibit greater compressibility, their structures naturally holding air which is pushed out by the consolidating pressure. Figure 7 shows the bulk density measurements of potato starch as a function of applied consolidating pressure which were made using the instrument's standard bulk compressibility test.

Figure 7: Compressibility data for potato flour



Calculating the design parameters

Hopper half angle, and subsequently μ , are calculated directly from wall friction angle and internal angle of friction, either graphically or using (complex) equations, and found to be 15° and 1.35 respectively. Imposing the μ plot on the existing FF chart gives the intersect value required for the calculation of the stress in the arch at the no flow/flow transition point (see figure 3). Outlet size is then calculated from the equation below.

$$B = \frac{\bar{\sigma}_1 H(\alpha)}{\rho g}$$

Where B is the outlet diameter (metres)
 $\bar{\sigma}_1$ is the consolidated stress generated in an arch at the outlet (kPa)
 $H(\alpha)$ is a function that takes account of variation in the arch thickness, hopper half angle & hopper geometric configuration
 ρ is the bulk density when consolidated at $\bar{\sigma}_1$ (kg/m^3 or g/ml)
 g is the acceleration due to gravity (9.81m/s^2)

At this point in the calculation, accuracy can be improved by assessing sensitivity to the value of internal angle of friction. The value of major consolidating stress at the point of intersection can be used to generate a more representative value of internal angle of friction compared with the original, averaged value (see figure 5c). This 'first iteration value', 45.5° in this case, enables the generation of a new half angle, μ , and arch stress. As subsequent iterations produce very little change feeding these figures into the equation above is acceptable. A hopper half angle of 15° is required and an outlet size of 0.59m.

As with all engineering design the final results are modified to provide a margin for error. Standard practice is to decrease hopper half angle by 3° , i.e. make it steeper than the analysis suggests, and increase outlet size by 20%. This gives the following design parameters for this hopper:

Hopper half angle : 12°
 Outlet size : 0.71m

In conclusion

Smart software in combination with modern automated powder testers demystifies hopper design – from powder testing through to specification. It improves access to a well-established but complex design methodology allowing the majority of process engineers to independently and confidently specify critical hopper dimensions for any given powder. Since outsourcing hopper design, and the associated powder testing, is costly, this solution offers financial benefits, and simultaneously enhances the company's ability to successfully operate and utilise storage silos.

Working through the design process generates an understanding of which factors dominate and compromise operation. Furthermore, bringing testing in-house makes it easier to assess the sensitivity of a powder and hopper design to changes induced by, for example, segregation or a varying moisture level. The developed understanding allows a hopper to be confidently re-used for alternative purposes. It also enables a process engineer to effectively assess options for modifying the process or hopper to give optimal ongoing performance.

References

- [1] Storage and flow of solids, Bulletin 123 of the Utah Engineering Experiment Station, November 1964 (Revised 1980), A.W.Jenike, University of Utah.
- [2] Basic principles of bulk solids storage, flow and handling, A.W. Roberts, The Institute for Bulk Materials Handling Research, 1993
- [3] Standard Shear Testing Technique for particulate solids using the Jenike Shear cell. IChemE/EFCE, 198

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